

AFRICA: DIGITAL DEVELOPMENT & ECONOMIC PERFORMANCE

Final Data Analytics Capstone Report — World Development Indicators (WDI)

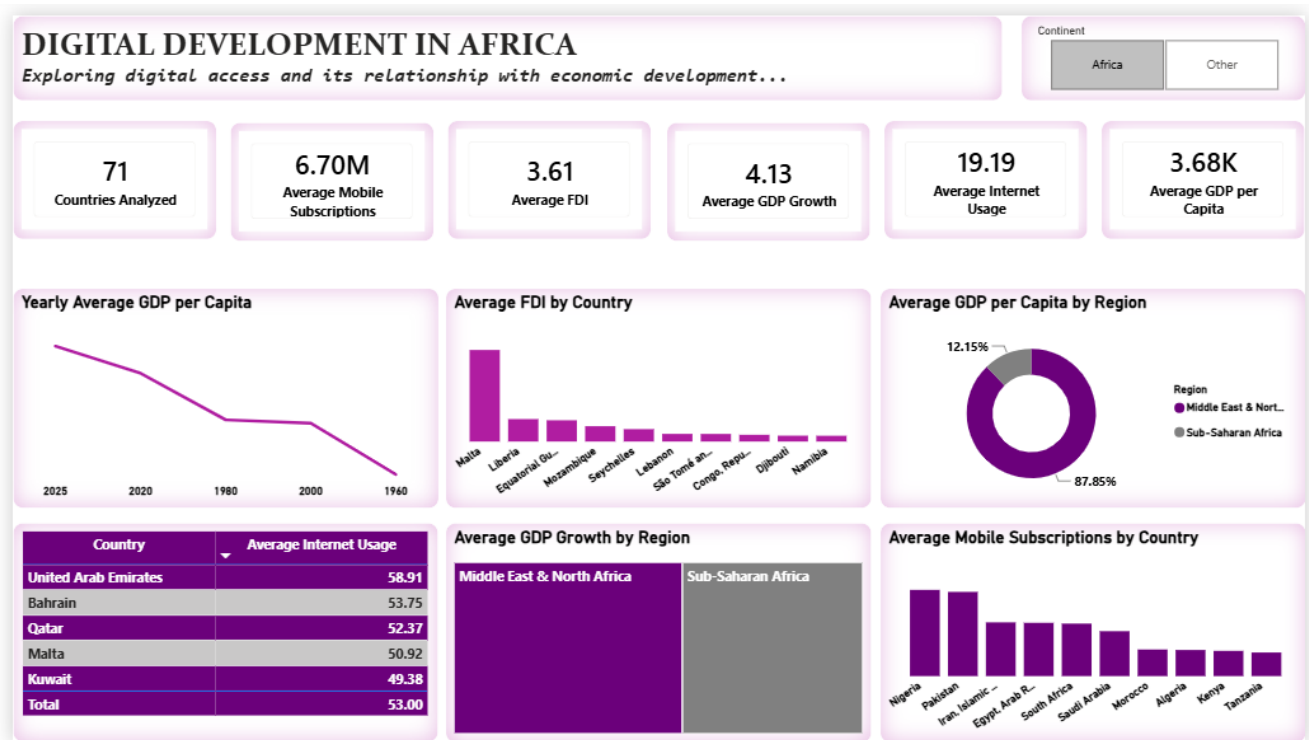


Figure 1. Final dashboard view supplied for the capstone report.

Executive Summary

This project examines the relationship between digital access, foreign direct investment (FDI), economic growth and GDP per capita using World Bank World Development Indicators. The dashboard combines country-level comparisons, regional summaries and yearly trends to provide a concise view of digital and economic development.

Key message: digital connectivity is an important development indicator, but the dashboard should be interpreted as descriptive evidence rather than proof that mobile or internet access directly causes higher FDI or economic growth.

1. Dashboard Snapshot

Metric	Value shown
Countries analyzed	71
Average mobile subscriptions	6.70M
Average FDI	3.61
Average GDP growth	4.13
Average internet usage	19.19
Average GDP per capita	3.68K

2. Key Findings

Digital access: The dashboard reports an average of 6.70 million mobile subscriptions and average internet usage of 19.19. These indicators provide a broad picture of digital connectivity across the countries represented.

Foreign direct investment: Average FDI is displayed as 3.61. The country chart shows substantial variation, indicating that FDI is not evenly distributed across the countries in the dataset.

Economic performance: Average GDP growth is shown as 4.13, while average GDP per capita is 3.68K. The dashboard therefore captures both short-run growth and a broader income-level measure.

Regional differences: The regional visuals distinguish Middle East & North Africa from Sub-Saharan Africa. This is useful because the World Bank's regional classification places North African countries within MENA rather than Sub-Saharan Africa.

Internet-use leaders shown: The visible country table shows the highest displayed internet-usage values led by United Arab Emirates (58.91), Bahrain (53.75), Qatar (52.37), Malta (50.92) and Kuwait (49.38).

3. Analytical Interpretation

The dashboard is designed to investigate whether countries with stronger digital connectivity also tend to show stronger economic outcomes. The scatter relationship between mobile subscriptions and FDI is especially useful for identifying clusters, outliers and broad associations. However, mobile subscriptions are not the same as unique users, and FDI can be affected by many factors including market size, natural resources, institutions, exchange rates, political conditions and trade policy. Consequently, the visuals support exploration and comparison rather than causal conclusions.

4. Data Preparation & Dashboard Design

The project uses World Development Indicators data and a country dimension containing World Bank country codes and regional classifications. The WDI data was transformed from its wide indicator-by-year structure into an analysis-friendly form, and a relationship was established between the country dimension and the main WDI table.

A custom **Continent** classification was created from the World Bank **Region** field so that both **Sub-Saharan Africa** and **Middle East & North Africa** are classified as **Africa**. This avoids excluding North African countries simply because the World Bank groups them under MENA.

5. Dashboard Components

Component	Purpose
KPI cards	Summarize countries, mobile subscriptions, FDI, GDP growth, internet usage and GDP per capita.
Yearly GDP per capita	Shows how the income-level indicator changes over time.
Average FDI by country	Highlights differences in FDI across countries.
GDP per capita by region	Compares the two African World Bank regions.
Internet usage table	Provides a ranked country-level view of internet usage.
GDP growth by region	Compares average growth across MENA and Sub-Saharan Africa.
Mobile subscriptions by country	Compares digital/mobile connectivity across countries.

6. Quality-Control Note

Before final submission, verify that the **Continent = Africa** filter is actually affecting every country-level visual. In the supplied screenshot, the visible internet-usage table includes countries such as the United Arab Emirates, Bahrain and Qatar even though the dashboard is titled for Africa. If the Africa selection is intended to be active, those values suggest that at least one visual may not be receiving the continent filter. This should be checked in Power BI before presenting the dashboard as an Africa-only analysis.

7. Recommendations

1. Verify the Africa slicer/filter interaction across every visual and confirm that all Africa-only visuals contain the intended countries.
2. Keep the World Bank Region field available so MENA and Sub-Saharan Africa can be compared without redefining the underlying source classification.
3. Use the scatter plot to identify outliers and investigate them individually rather than interpreting correlation as causation.
4. For future analysis, consider adding population, GDP size, fixed broadband, electricity access and urbanization to provide additional context.
5. Document the units and definitions of the FDI and internet-use indicators directly in the report or dashboard tooltips.

Conclusion

The completed dashboard provides a strong visual foundation for examining digital development and economic performance across African countries. Its main value is in making differences between countries and regions easy to

explore. After the Africa filter is validated across all visuals, the dashboard will be ready to serve as a concise portfolio demonstration of data preparation, Power BI modeling, KPI design, comparative analysis and data storytelling.

Data source: World Bank — World Development Indicators (WDI). Dashboard values and visuals in this report are based on the supplied final Power BI screenshot.